season_solar_btu = 7280153.54
solar_share_percent = 66.66
fuel_units_needed = 0.18
sources: fw.nyserda_gasifier (no evidence given); sip.prs610_panel_thickness (published); sip.prs610_core_thickness (published); osb.gp_7_16 (published); sip.sipa_definition (published); wt.syn_osb_sheet (inferred); e1295.core (published); ev.sip_insulspan_4half_eps_r (published); sip.premier_thicknesses (published); pd.esr4524_facing (published); osb.gp_7_16 (inferred); pd.esr4524_core (published); e1295.core (inferred); wd.door_thermatru_smoothstar_120 (published); ev.door_thermatru_fc_ss_opaque_u (published); wd.vitro_td140_igu_65psf (published); ev.igu.viracon_u (published); ev.igu.viracon_shgc (published); ev.igu.viracon_vt (published); site.frost_mrc (published); site.snow_kent (published); site.live_sleeping (published); site.dead_ten_psf (published); code.r311_2_door (published); rf.noaa_hdd (published); wo.pvcalc_south_oct_apr (inferred); wo.pvcalc_east_oct_apr (inferred); wo.pvcalc_west_oct_apr (inferred); wo.pvcalc_north_oct_apr (inferred); ev.wood.msu_hardwood_cord (published)
joist-deflection (structural): known
deflection_mm = 2.88
span_over_deflection = 767.49
bending_stress_mpa = 3.82
sources: lum.nds.fv_spf_no2 (visual-read); lum.nds.cf_2x6 (visual-read); lum.nds.cr (published); wd.awc_wsdd_spf_vr (visual-read); site.frost_mrc (published); site.snow_kent (published); site.live_sleeping (published); site.dead_ten_psf (published); code.r311_2_door (published); rf.noaa_hdd (published); wo.pvcalc_south_oct_apr (inferred); wo.pvcalc_east_oct_apr (inferred); wo.pvcalc_west_oct_apr (inferred); wo.pvcalc_north_oct_apr (inferred)
gravity-load-path (structural): known
each contact under a piece takes a share of its force in proportion to the contact's area: an approximation that ignores where the load sits on the piece and, for a piece with more supports than it needs, how stiff each support is; an opening hangs in the hosts it is cut from by the cut's volume, and a hung piece by the faces at its sides
pieces = 27
pieces_weight_known = 27
pieces_force_known = 27
ground_reaction_n = 28007.33
sources: site.frost_mrc (published); site.snow_kent (published); site.live_sleeping (published); site.dead_ten_psf (published); code.r311_2_door (published); rf.noaa_hdd (published); wo.pvcalc_south_oct_apr (inferred); wo.pvcalc_east_oct_apr (inferred); wo.pvcalc_west_oct_apr (inferred); wo.pvcalc_north_oct_apr (inferred); sip.prs610_panel_thickness (published); sip.prs610_core_thickness (published); osb.gp_7_16 (published); sip.sipa_definition (published); wt.syn_osb_sheet (inferred); e1295.core (published); ev.sip_insulspan_4half_eps_r (published); sip.premier_thicknesses (published); pd.esr4524_facing (published); osb.gp_7_16 (inferred); pd.esr4524_core (published); e1295.core (inferred); lum.nds.fv_spf_no2 (visual-read); lum.nds.cf_2x6 (visual-read); lum.nds.cr (published); wd.awc_wsdd_spf_vr (visual-read); wd.advantech_23_32_commerce (published); wd.advantech_sheet_size (published); wd.vitro_td140_igu_65psf (published); ev.igu.viracon_u (published); ev.igu.viracon_shgc (published); ev.igu.viracon_vt (published); wd.door_thermatru_smoothstar_120 (published); ev.door_thermatru_fc_ss_opaque_u (published); wt.dressed_sizes (published); fo.post.life_uc4b_ret023 (published); wt.syn_post_6x6 (inferred)
header-check (structural): known
reference design values with no NDS adjustment factor; the roof carries the ground snow load unreduced; a header's load is the roof over its wall and the joists landing at its head, not its own weight nor the wall or glass above it
headers = 3
headers_over_one = 0
utilisation_max = 0.41
sources: site.frost_mrc (published); site.snow_kent (published); site.live_sleeping (published); site.dead_ten_psf (published); code.r311_2_door (published); rf.noaa_hdd (published); wo.pvcalc_south_oct_apr (inferred); wo.pvcalc_east_oct_apr (inferred); wo.pvcalc_west_oct_apr (inferred); wo.pvcalc_north_oct_apr (inferred); lvl.width (published); lvl.depth (published); gl.prop.lvl_2_0e_2600 (visual-read); gl.prop.lvl_density_unpublished (unpublished); wt.lvl_weyerhaeuser_244 (visual-read)
work-at-published-rates (tasks): known
post_holes_hours = 2
set_posts_hours = 2
floor_hours = 1
walls_hours = 5.63
roof_hours = 1.91
cut_openings_hours = 6
windows_and_door_hours = 14
frame_loft_hours = 7.2
lay_subfloor_hours = 3
install_boiler_hours = 16
measured_hours = 58.74
listed_usd = 8539.91
pieces_not_priced = 0
not_measured =
sources: task.t1.hours_hand_ramirent (no evidence given); task.t5.hours_per_sqft_range (no evidence given); task.t8.hours_per_sqft_roof (no evidence given); hr.t2.set_post_hours_unpublished (no evidence given); hr.t6.sip_field_opening_hours_unpublished (no evidence given); hr.t7.window_forum_table (no evidence given); hr.t7.door_range_inferred (no evidence given); task.t3.hours_per_sqft_joists (no evidence given); hr.t4.subfloor_hours_unpublished (no evidence given); hr.t11.boiler_hydronic_total_hours_unpublished (no evidence given); pr1.post6x6x8.treated.chelsea_low (no evidence given); pr1.joist2x6x8.untreated.chelsea_low (no evidence given); pr1.subfloor.advantech_tg.chelsea_low (no evidence given); pr2.door.superarbor_9lite_fg (no evidence given); pr2.window.fixr_4x4_low (no evidence given); pr2.window.fixr_4x8_low (no evidence given); pr3.boiler.econoburn_eko_about_5000 (no evidence given); sp.acme_other_thicknesses (published); sp.null_michigan_maker_price (unpublished); sp.acme_1025_price (published)

RULES, EACH WITH ITS STATUS

posts-under-floor [foundation]: PASS: every floor corner has a post under it; source: integrity: a floor on posts needs a post at each corner
walls-enclose-floor [walls]: PASS: walls close every edge of the floor; source: integrity: a shell closes every edge of its floor
roof-is-over-floor [roof]: PASS: a roof stands over the whole floor; source: integrity: a roof stands over the floor it keeps dry
egress-door-size [doors]: PASS: the egress door is at least as large as the limit; source: Michigan Residential Code 2015 R311.2, research row code.r311_2_door: side-hinged, 3 ft by 6 ft 8 in
posts-below-frost-line [foundation]: PASS: the posts are buried below the frost line; source: Michigan Residential Code R403.1.4: footings 42 in below grade, the site's frost_depth_in (research row site.frost_mrc)
roof-covers-floor [roof]: PASS: the roof covers the floor; source: integrity: a roof keeps the floor under it dry
headers-within-capacity [header_check]: PASS: every header is within its capacity; source: the header's catalog design values: bending, shear and its deflection limit
joists-span-the-deck [joists]: PASS: the joists span the deck at the stated spacing; source: integrity: a deck needs joists from edge to edge at the stated spacing

GRAVITY LOAD PATH, PER POST

status: known; each contact under a piece takes a share of its force in proportion to the contact's area: an approximation that ignores where the load sits on the piece and, for a piece with more supports than it needs, how stiff each support is; an opening hangs in the hosts it is cut from by the cut's volume, and a hung piece by the faces at its sides
post@2908,2438,-1219~140x140: ground reaction 7001.83 N
post@2908,4736,-1219~140x140: ground reaction 7001.83 N
post@610,2438,-1219~140x140: ground reaction 7001.83 N
post@610,4736,-1219~140x140: ground reaction 7001.83 N

HEADER UTILISATIONS

status: known; largest 0.41
door@610,4819: utilisation 0.02; bending 0.19 MPa, shear 0.05 MPa, deflection 0.01 mm; load 1.4 N/mm against capacity 57.63 N/mm
glazing@610,2495,1219: utilisation 0; bending 0 MPa, shear 0 MPa, deflection 0 mm; load 0 N/mm against capacity 28.82 N/mm
glazing@610,2495,2438: utilisation 0.41; bending 6.47 MPa, shear 0.8 MPa, deflection 1.93 mm; load 11.73 N/mm against capacity 28.82 N/mm

BUILD SIMULATION TOTALS

ended quiescent; clock step 1 minute until 4800
finish_time = 33.22
labour_hours = 61.8
learning = 0.58
idle = 32
score = -150.39
final stocks:
boiler_installed = 1
budget_usd = 5512.59
cost_usd = 14487.41
deck_sheets_laid = 2
floor_panel_ft2_set = 63.98
joists_set = 7
openings_cut = 3
openings_set = 3

PROJECT	SHEET	SHEET NUMBER	DATE
Small house on posts, from Jim's pencil sketch	**Design summary**	S-1 (page 2 of 3)	2026-10-20
CONFIG COMMIT	ENGINE PIN	SCALE	SOURCE
31254d4	we 5af4c5f; rendered with we c3542b1	not to scale	we geometry